EXPERIMENT 15

AIM

Monitor a directory for a specified period and record created, modified and deleted files with timestamps.

ALGORITHM

1. Specify the directory to monitor.

2. Take an initial file snapshot.

3. Wait for the observation interval.

4. Take a second snapshot.

5. Find created and deleted files by comparing filenames.

6. Find modified files by comparing timestamps.

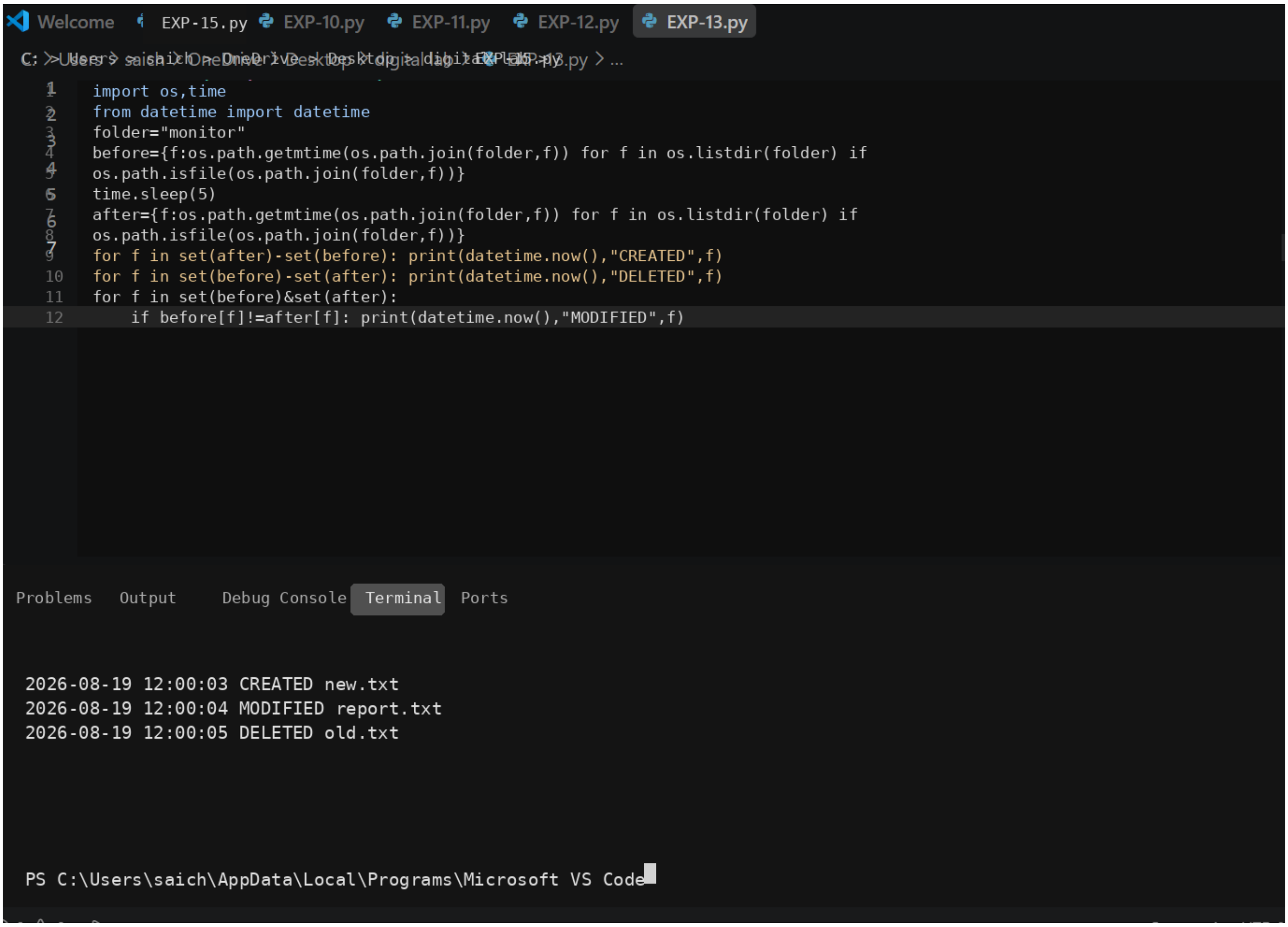
7. Display each activity with a timestamp.

8. Stop.

CODE

```
import os
import time
from datetime import datetime

folder = "monitor"

def snapshot():
    result = {}
    for name in os.listdir(folder):
        path = os.path.join(folder, name)
        if os.path.isfile(path):
            result[name] = os.path.getmtime(path)
    return result

before = snapshot()
time.sleep(5)
after = snapshot()

for name in sorted(set(after) - set(before)):
    print(datetime.now(), "CREATED", name)

for name in sorted(set(before) - set(after)):
    print(datetime.now(), "DELETED", name)

for name in sorted(set(before) & set(after)):
    if before[name] != after[name]:
        print(datetime.now(), "MODIFIED", name)
```

OUTPUT



RESULT

The program records observable file-system changes and produces an activity summary.